

GED Science Practice Test 8

PART A — QUESTIONS

GED-SCI8-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student investigates how salt concentration affects freshwater single-celled organisms. Four samples contain the same number of organisms and are kept at the same temperature and light level. Each sample is placed in water with a different salt concentration. After 30 minutes, the student counts the number of organisms still moving normally.

Salt concentration (%) | Organisms moving normally

0.0 | 48

0.2 | 43

0.6 | 19

1.0 | 6

Which conclusion is best supported by the data?

- A) Higher salt concentration reduced the number of organisms moving normally.
- B) Higher salt concentration increased normal movement in all samples.
- C) Salt concentration had no relationship to organism movement.
- D) All organisms moved normally at 1.0% salt.

GED-SCI8-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the work done when pushing different carts. Work can be calculated as force $\times$ distance.

Cart | Force (N) | Distance (m)

A | 25 | 4

B | 40 | 2

C | 15 | 8

D | 60 | 1

Which cart requires the greatest work?

- A) Cart A
- B) Cart B
- C) Cart C
- D) Cart D

GED-SCI8-003

Section: Science

Type: mcq

Category: Earth and Space Science
Skill: Reading for Meaning in Science
Prompt:

A planet's axis is tilted as it orbits its star. During part of the orbit, one hemisphere is tilted toward the star and receives more direct sunlight for longer daily periods. During the opposite part of the orbit, that hemisphere is tilted away and receives less direct sunlight.

Which phenomenon is mainly explained by this model?

- A) Seasons
- B) Moon phases
- C) Earthquakes
- D) Mineral hardness

GED-SCI8-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The small intestine has many folds and tiny fingerlike projections called villi. These structures increase the surface area available for absorbing digested nutrients into the blood. Nutrients then travel to body cells where they can be used for energy, growth, and repair.

What is the main advantage of villi in the small intestine?

- A) They increase the surface area for nutrient absorption.
- B) They pump blood through the heart.
- C) They produce oxygen for the lungs.
- D) They store inherited information in chromosomes.

GED-SCI8-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A machine lifts a 90 N load a distance of 3 meters.

$\text{Work} = \text{force} \times \text{distance}$

How much work is done in joules?

Type your answer as a whole number.

GED-SCI8-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student tests how the length of a wire affects the brightness of a bulb in a circuit. The student uses wires of different lengths but keeps the wire material and battery the same.

Which TWO factors should be kept constant?

- A) Wire material
- B) Battery used
- C) Wire length
- D) Bulb brightness
- E) Final measured result

GED-SCI8-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Mitochondrial DNA is usually inherited from the egg cell. In a family, a mother with a mitochondrial DNA condition may pass it to [D1], while an affected father usually [D2].

Dropdowns:

D1) her children | only unrelated neighbors | only plants

D2) does not pass mitochondrial DNA to his children | passes it to all children through sperm mitochondria | changes it into a protein

GED-SCI8-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student pours vinegar into a solution containing a few drops of red cabbage indicator. The solution turns pink. Red cabbage indicator turns pink in acidic solutions, purple in neutral solutions, and green-blue in basic solutions.

What can the student conclude?

- A) The vinegar solution is acidic.
- B) The vinegar solution is strongly basic.
- C) The vinegar solution contains no particles.
- D) The indicator prevents chemical testing.

GED-SCI8-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A city plans to reduce summer heat in neighborhoods with many paved surfaces. One proposal is to plant more trees. Trees provide shade and release water vapor from leaves, which can cool nearby air. Another proposal is to replace dark roofs with lighter roofs that reflect more sunlight.

Which outcome would best support these proposals?

- A) Neighborhood surface temperatures decrease after adding trees and lighter roofs.

- B) Neighborhood temperatures increase after shade is added.
- C) Lighter roofs absorb all incoming sunlight.
- D) Trees prevent all weather changes from occurring.

GED-SCI8-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A doctor records blood oxygen saturation for four patients.

Patient | Blood oxygen saturation (%)

A | 98

B | 93

C | 89

D | 96

Which patient has the lowest blood oxygen saturation?

- A) Patient A
- B) Patient B
- C) Patient C
- D) Patient D

GED-SCI8-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

A) Angle of a mirror

B) Direction of reflected light

C) Same incoming light source

D) Color of the table

Targets:

T1) Independent variable

T2) Dependent variable

T3) Controlled variable

GED-SCI8-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A neuron sends signals using electrical impulses and chemical messengers. The long axon carries an impulse away from the cell body. At the end of the neuron, chemical messengers can cross a small gap and pass the signal to another cell.

Which statement best describes one function of neurons?

- A) They transmit information through the nervous system.
- B) They form mineral crystals in rocks.
- C) They filter sediment from rivers.
- D) They reflect sunlight from planets.

GED-SCI8-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A sealed flexible bag contains air. When the bag is placed in a freezer, the air inside cools and the bag shrinks slightly. Cooling causes gas particles to move more slowly and collide with the bag walls less forcefully.

Which statement best explains the shrinking bag?

- A) Cooling the gas lowered its pressure inside the flexible bag.
- B) Cooling the gas created new gas particles.
- C) The gas changed into a metal.
- D) The bag shrank because gases have no particles.

GED-SCI8-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A town installs 45 rain gardens. During one storm, the rain gardens capture a total of 9,000 liters of runoff.

What is the average amount of runoff captured per rain garden?

Type your answer as a whole number.

GED-SCI8-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether a probiotic supplement affects digestion time in volunteers. One group takes the supplement each day, while a similar group takes a capsule with no active ingredient. Both groups eat the same prepared meals during the study. Digestion time is recorded for each volunteer.

Why is the capsule with no active ingredient included?

- A) It provides a placebo comparison group.
- B) It guarantees that digestion will stop.
- C) It changes the supplement into food.
- D) It removes the need to record data.

GED-SCI8-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student compares how different surfaces reflect light. A flashlight shines at the same angle onto aluminum foil, white paper, black cloth, and cardboard. A light sensor measures reflected light from each surface.

What is the dependent variable?

- A) Amount of reflected light measured by the sensor
- B) Type of surface
- C) Flashlight angle
- D) Distance between the flashlight and surface

GED-SCI8-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

In a transverse wave, the material moves perpendicular to the direction the wave travels. The highest point of the wave is the [D1], and the distance from one crest to the next crest is the [D2].

Dropdowns:

D1) crest | trough | circuit

D2) wavelength | mass | voltage

GED-SCI8-018

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A technician measures blood pressure before and after a person sits quietly for 10 minutes.

Measurement | Before rest | After rest

Systolic pressure (mm Hg) | 138 | 124

Diastolic pressure (mm Hg) | 86 | 78

Which statement is best supported by the data?

- A) Both systolic and diastolic pressure decreased after rest.
- B) Both systolic and diastolic pressure increased after rest.
- C) Blood pressure stayed exactly the same after rest.
- D) Diastolic pressure increased while systolic pressure decreased.

GED-SCI8-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the current in a circuit with different resistors while using the same battery.

Resistance (Ω): 2, 4, 6, 8

Current (A): 6, 3, 2, 1.5

Which statement is best supported by the data?

- A) Current decreases as resistance increases.
- B) Current increases as resistance increases.
- C) Current stays the same for all resistors.
- D) Resistance has no relationship to current.

GED-SCI8-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A county tracks electricity production from two renewable sources.

Month | Wind energy (MWh) | Solar energy (MWh)

April | 310 | 240

May | 280 | 315

June | 250 | 410

July | 220 | 455

Which TWO statements are best supported by the data?

- A) Solar energy production increased from April to July.
- B) Wind energy production decreased from April to July.
- C) Wind energy production increased every month.
- D) Solar energy production was lowest in July.
- E) Both energy sources produced exactly the same amount each month.

GED-SCI8-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person's skin helps maintain body temperature. When the body becomes too warm, blood vessels near the skin can widen, allowing more heat to leave the body. Sweat glands can also release sweat, which cools the skin as it evaporates.

Which body system function is described?

- A) Regulation and protection by the integumentary system
- B) Digestion of food by the stomach
- C) Production of sperm by the reproductive system
- D) Formation of igneous rock

GED-SCI8-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student compares two samples. Sample 1 contains only copper atoms. Sample 2 contains water molecules, each made of hydrogen and oxygen atoms chemically bonded together.

Which statement correctly classifies the samples?

- A) Copper is an element, and water is a compound.
- B) Copper is a compound, and water is an element.
- C) Both copper and water are elements.
- D) Both copper and water are mixtures.

GED-SCI8-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A geologist finds a rock with crystals arranged in bands. The rock formed when an older rock was changed by high heat and pressure deep underground, but it did not completely melt.

Which type of rock is described?

- A) Metamorphic rock
- B) Igneous rock formed from lava
- C) Sedimentary rock made from loose sand
- D) A meteorite from space

GED-SCI8-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A survey of 180 adults finds that 63 adults report sleeping fewer than 7 hours per night.

What percent of the adults report sleeping fewer than 7 hours per night?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI8-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests whether adding salt changes the boiling point of water. Four beakers each contain 200 mL of water. Different amounts of salt are added to three of the beakers, and one beaker has no

salt. The student heats each beaker on the same hot plate setting and records the boiling temperature.

Which setup best improves the investigation?

- A) Repeat each salt amount in several trials.
- B) Use a different water volume in every beaker.
- C) Change the hot plate setting for each beaker.
- D) Record only the highest boiling temperature.

GED-SCI8-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A cell copies its DNA before dividing. The copied chromosomes are then separated so each new body cell receives a complete set of genetic instructions.

Which process is mainly described?

- A) Mitosis
- B) Fertilization
- C) Photosynthesis
- D) Evaporation

GED-SCI8-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares four lamps.

Lamp | Energy used in 1 hour (kJ) | Light output rating

- A | 120 | 8
- B | 90 | 8
- C | 150 | 8
- D | 200 | 8

If all lamps give the same light output rating, which lamp is most energy efficient?

- A) Lamp A
- B) Lamp B
- C) Lamp C
- D) Lamp D

GED-SCI8-028

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records the apparent brightness of four stars as seen from Earth. Smaller magnitude numbers mean brighter stars.

Star | Apparent magnitude

A | 1.2

B | 3.5

C | -0.4

D | 5.1

Which star appears brightest from Earth?

A) Star A

B) Star B

C) Star C

D) Star D

GED-SCI8-029

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A public health worker records the average number of mosquito larvae in standing water containers before and after lids are added.

Condition | Average mosquito larvae per container

Open containers | 34

Covered containers | 5

Which claim is best supported by the data?

A) Covering containers was associated with fewer mosquito larvae.

B) Covering containers increased mosquito larvae.

C) Mosquito larvae were absent from open containers.

D) Container lids had no relationship to mosquito larvae.

GED-SCI8-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student mixes two clear liquids. The mixture becomes cloudy and a solid forms at the bottom of the test tube. The solid does not dissolve when the tube is gently shaken.

What evidence suggests that a chemical reaction occurred?

A) A new solid precipitate formed.

B) The liquids stayed completely unchanged.

C) The test tube became invisible.

D) The liquids separated only because of gravity.

GED-SCI8-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A researcher studies whether screen brightness before bedtime affects melatonin levels. Volunteers use a tablet at low, medium, or high brightness for 30 minutes before sleep. Melatonin level is measured afterward.

Which TWO steps would improve the investigation?

- A) Keep tablet use time the same for all volunteers.
- B) Measure melatonin using the same method for all volunteers.
- C) Let each volunteer choose a different tablet-use time.
- D) Record only the results from the high-brightness group.
- E) Change brightness and room temperature together.

GED-SCI8-032

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A toy motor lifts a 20 N weight in different amounts of time. The same work is done in each trial.

Trial | Work done (J) | Time (s)

- A | 100 | 5
- B | 100 | 10
- C | 100 | 20
- D | 100 | 25

Which trial has the greatest power output?

- A) Trial A
- B) Trial B
- C) Trial C
- D) Trial D

GED-SCI8-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Some bacteria live in nodules on the roots of bean plants. The bacteria convert nitrogen gas into forms the plant can use. The plant provides the bacteria with sugars and a protected place to live.

Which statement best describes this relationship?

- A) Both organisms benefit from the relationship.
- B) The bacteria benefit, but the plant is always harmed.
- C) The plant benefits, but the bacteria receive nothing.
- D) Neither organism is affected.

GED-SCI8-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person eats a meal high in fat. Bile released into the small intestine breaks large fat droplets into smaller droplets. This increases the surface area available for digestive enzymes to act on the fat.

What is the main role of bile in this process?

- A) It breaks large fat droplets into smaller droplets to aid digestion.
- B) It carries oxygen from the lungs to body cells.
- C) It stores chromosomes before cell division.
- D) It forms hard minerals in bones.

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI8-001 — Correct: A — Correct Answer: Higher salt concentration reduced the number of organisms moving normally. — Explanation: The number of organisms moving normally decreased from 48 to 6 as salt concentration increased.

GED-SCI8-002 — Correct: C — Correct Answer: Cart C — Explanation: Work = force $\times$ distance. Cart C requires $15\text{ N} \times 8\text{ m} = 120\text{ J}$, the greatest amount listed.

GED-SCI8-003 — Correct: A — Correct Answer: Seasons — Explanation: Seasons are mainly caused by a planet's axial tilt as it orbits its star.

GED-SCI8-004 — Correct: A — Correct Answer: They increase the surface area for nutrient absorption. — Explanation: Villi increase the small intestine's surface area, allowing more efficient nutrient absorption.

GED-SCI8-005 — Correct: 270 — Tolerance: exact — Correct Answer: 270 — Explanation: Work = $90\text{ N} \times 3\text{ m} = 270\text{ J}$.

GED-SCI8-006 — Correct: A,B — Correct Answer: Wire material | Battery used — Explanation: Wire material and battery should remain constant so wire length is the main tested variable.

GED-SCI8-007 — Correct: D1=her children; D2=does not pass mitochondrial DNA to his children — Correct Answer: D1=her children; D2=does not pass mitochondrial DNA to his children — Explanation: Mitochondrial DNA is usually inherited from the egg, so mothers may pass it to children, while fathers usually do not.

GED-SCI8-008 — Correct: A — Correct Answer: The vinegar solution is acidic. — Explanation: Red cabbage indicator turning pink indicates an acidic solution.

GED-SCI8-009 — Correct: A — Correct Answer: Neighborhood surface temperatures decrease after adding trees and lighter roofs. — Explanation: Shade, water vapor release from leaves, and greater reflection from light roofs can reduce surface heating.

GED-SCI8-010 — Correct: C — Correct Answer: Patient C — Explanation: Patient C has the lowest listed oxygen saturation at 89%.

GED-SCI8-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Angle of a mirror; T2=Direction of reflected light; T3=Same incoming light source — Explanation: The mirror angle is changed, the reflected-light direction is measured, and the incoming light source is controlled.

GED-SCI8-012 — Correct: A — Correct Answer: They transmit information through the nervous system. — Explanation: Neurons transmit signals using electrical impulses and chemical messengers.

GED-SCI8-013 — Correct: A — Correct Answer: Cooling the gas lowered its pressure inside the flexible bag. — Explanation: Cooler gas particles move more slowly and collide less forcefully, lowering pressure and allowing the bag to shrink.

GED-SCI8-014 — Correct: 200 — Tolerance: exact — Correct Answer: 200 — Explanation: $9,000 \text{ liters} \div 45 \text{ rain gardens} = 200 \text{ liters per rain garden}$.

GED-SCI8-015 — Correct: A — Correct Answer: It provides a placebo comparison group. — Explanation: A placebo group helps researchers compare the supplement group with a similar group that did not receive the active supplement.

GED-SCI8-016 — Correct: A — Correct Answer: Amount of reflected light measured by the sensor — Explanation: The dependent variable is the measured result after changing the surface type.

GED-SCI8-017 — Correct: D1=crest; D2=wavelength — Correct Answer: D1=crest; D2=wavelength — Explanation: The crest is the highest point of a transverse wave, and wavelength is the distance between matching points such as crest to crest.

GED-SCI8-018 — Correct: A — Correct Answer: Both systolic and diastolic pressure decreased after rest. — Explanation: Systolic pressure decreased from 138 to 124 mm Hg, and diastolic pressure decreased from 86 to 78 mm Hg.

GED-SCI8-019 — Correct: A — Correct Answer: Current decreases as resistance increases. — Explanation: The table shows current decreasing from 6 A to 1.5 A as resistance increases from 2Ω to 8Ω .

GED-SCI8-020 — Correct: A,B — Correct Answer: Solar energy production increased from April to July. | Wind energy production decreased from April to July. — Explanation: Solar production rose from 240 to 455 MWh, while wind production fell from 310 to 220 MWh.

GED-SCI8-021 — Correct: A — Correct Answer: Regulation and protection by the integumentary system — Explanation: Skin helps regulate body temperature and protects the body.

GED-SCI8-022 — Correct: A — Correct Answer: Copper is an element, and water is a compound. — Explanation: Copper contains one type of atom, while water contains hydrogen and oxygen chemically bonded.

GED-SCI8-023 — Correct: A — Correct Answer: Metamorphic rock — Explanation: Metamorphic rocks form when existing rocks are changed by heat and pressure without completely melting.

GED-SCI8-024 — Correct: 35 — Tolerance: exact — Correct Answer: 35 — Explanation: $63 \div 180 \times 100 = 35\%$.

GED-SCI8-025 — Correct: A — Correct Answer: Repeat each salt amount in several trials. — Explanation: Repeating trials improves reliability and reduces the effect of random error.

GED-SCI8-026 — Correct: A — Correct Answer: Mitosis — Explanation: Mitosis produces body cells with complete sets of genetic instructions.

GED-SCI8-027 — Correct: B — Correct Answer: Lamp B — Explanation: Since all lamps give the same light output, the lamp using the least energy, Lamp B, is most efficient.

GED-SCI8-028 — Correct: C — Correct Answer: Star C — Explanation: Lower apparent magnitude means greater apparent brightness. Star C has the lowest value, -0.4.

GED-SCI8-029 — Correct: A — Correct Answer: Covering containers was associated with fewer mosquito larvae. — Explanation: Covered containers had an average of 5 larvae compared with 34 in open containers.

GED-SCI8-030 — Correct: A — Correct Answer: A new solid precipitate formed. — Explanation: Formation of a new solid precipitate is evidence that a chemical reaction occurred.

GED-SCI8-031 — Correct: A,B — Correct Answer: Keep tablet use time the same for all volunteers. | Measure melatonin using the same method for all volunteers. — Explanation: Keeping time and measurement method constant improves control and reliability.

GED-SCI8-032 — Correct: A — Correct Answer: Trial A — Explanation: $\text{Power} = \text{work} \div \text{time}$. Trial A has $100 \text{ J} \div 5 \text{ s} = 20 \text{ W}$, the greatest power output.

GED-SCI8-033 — Correct: A — Correct Answer: Both organisms benefit from the relationship. — Explanation: The bacteria receive sugars and protection, while the plant receives usable nitrogen.

GED-SCI8-034 — Correct: A — Correct Answer: It breaks large fat droplets into smaller droplets to aid digestion. — Explanation: Bile emulsifies fats, increasing surface area for digestive enzymes.